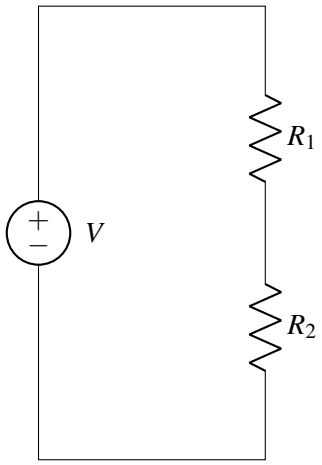
CSM
Fall 2026

EE 64
Worksheet 1

1. Voltage Divider Properties

Let's take a systematic look at the voltages across a resistor, and see how other components in the circuit can affect it. Consider the following circuit:



(a) Calculate the voltage drop across R_1 and R_2 using series resistance calculations.

Solution: The current out of the voltage source is given by Ohm's law:

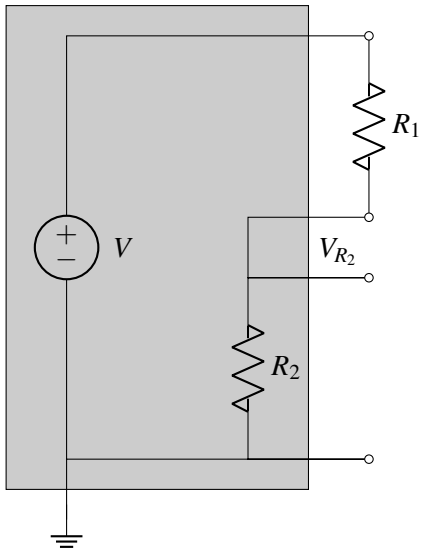
$$i = \frac{V}{R_{eq}}$$
$$i = \frac{V}{R_1 + R_2}$$

Again, by Ohm's law, we have that

$$V_{R_1} = iR_1 = V\frac{R_1}{R_1 + R_2}$$
$$V_{R_2} = iR_2 = V\frac{R_2}{R_1 + R_2}$$

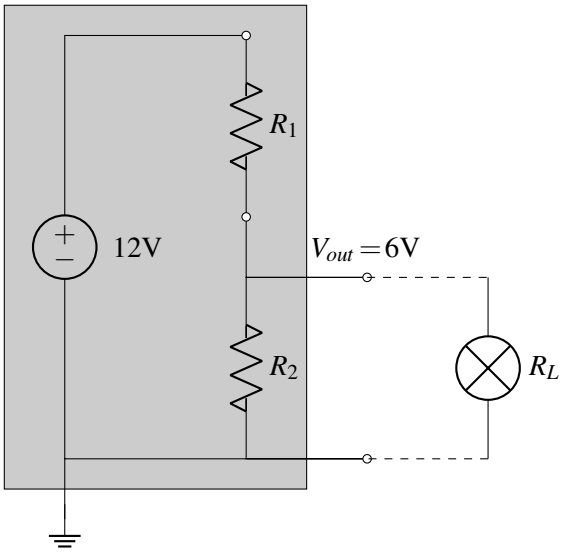
(b) Suppose we want to manipulate the voltage across R_2, but it's locked in a box with the voltage source, as denoted below. Can we use R_1 to manipulate V_{R_2}? What range of voltages can we achieve?

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Solution: Any voltage in the range $(0, V]$! Notice from the equations above that $V_{R_2} = V\frac{R_2}{R_{Total}}$. If we increase R_1 indefinitely, holding R_2 constant, we can make the fraction arbitrarily small. Intuitively, since the same current flows through both R_1 and R_2, they have to split the total voltage of the power source, and larger resistances correspond to larger voltage drops (by Ohm's Law). If we decrease R_1 to 0, $V_{R_2} = V$, so the voltage can be at most whatever is supplied by the power source. That the voltage source limits the achievable voltage in the circuit is a concept we will see again when we cover clipping in op-amps.

(c) Now let's try using our new variable voltage source to power a light bulb with resistance R_L, where the threshold voltage for lighting the bulb is 6V. Find R_1 and R_2 so that the voltage across R_2 is this threshold voltage; that is, $V_{R_2} \equiv V_{out} = 6\text{V}$. Assume we have a 12V voltage source.



Solution: We want to split the voltage in half (from 12 to 6). Based on the voltage divider formula above, that means $R_1 = R_2 \equiv R$. Note that, under this condition, the voltage is evenly split regardless of what the actual resistance values are! While the current depends on actual resistance values, the voltage only depends on the ratio of resistances.

(d) Now that we found an R_1 and R_2 that seem to divide our voltage source appropriately, let's try to

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connect the bulb to the ends of R_2. Remember, the bulb has a resistance R_L. Calculate the voltage across R_1, R_2 and the light bulb when it is connected. Will the light bulb turn on?

Solution: Let's reapply the voltage divider formula, but now notice that the "second" resistor is $R_2 \| V_L$!. When we change the resistance in a voltage divider circuit, the voltage across that resistance changes as well. We find that $V_{R_1} = V\frac{R}{R+R\|R_L} = 12\text{V}\frac{R}{R+\frac{RR_L}{R+R_L}} = 12\text{V}\frac{R+R_L}{R+2R_L}$. The rest of the potential drop must be across the R_2-R_L system, so we have $V_{R_2} = V_{\text{bulb}} = 12\text{V}\left(1-\frac{R+R_L}{R+2R_L}\right) = 12\text{V}\frac{R_L}{R+2R_L} \leq 12\text{V}\frac{R_L}{2R_L} = 6\text{V}$.

So the voltage *decreased*, and the light bulb doesn't turn on.

The takeaway from this is that, while it may seem like the voltage divider can make voltage sources of arbitrary voltages, the act of connecting a component actually changes the output of the voltage divider. We will later learn about a way to stop a light bulb (or other devices that use up power) from "affecting" the circuit that supplies power by placing a "buffer" between the two.

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2. Never fail at resistor equivalence

Solution: Description: derive intuition behind resistor equivalences. Learn about nodes and how to combine resistors in a real problem.

(a) Using what you learned about resistance above ($R = \frac{\rho \cdot l}{A}$), explain what would happen if you connect two resistors in series.

Solution: Make sure to draw out resistors as boxes in series and parallel for parts (a) and (b) respectively. The answer only makes sense once it is drawn out.

Solution: Intuitively, the amount of impedance the current faces by going through the two resistors in series would just be the sum of the impedance faced by the current going through each individual of the resistor. Thus, for n resistors in series,

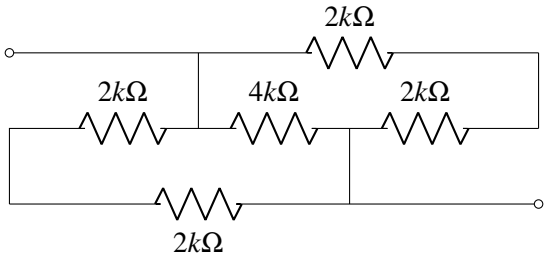
$$R_{eq} = R_1 + R_2 + \cdots + R_n$$

(b) Using the same resistance equation from part (a), explain what would happen if you connect two resistors in parallel.

Solution: Having resistors in parallel is like taking taking two boxes and combining them via their sides. This increases the cross-sectional area in the front. So intuitively, the equivalent resistance of resistors in parallel would be smaller than the resistance of the original resistors. In addition, intuitively, more current would choose to go to the side with less resistance. Thus, for n resistors in parallel,

$$R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}}$$

(c) Now let's apply some of these ideas to a circuit. For the circuit below, mark all the nodes.



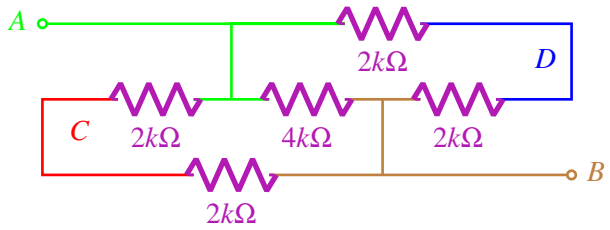
Solution: Mentors: at this point, explain the concept of nodes. Students may be confused as to what constitutes a node and what the properties of a node are.

Here is how I like to teach it: keep a pencil (marker) at a wire, and keep going in all directions till you hit any components. Having multiple colors of markers helps a ton here! Otherwise draw little symbols. Forward slashed lines on one node, backward slashed on another, circles on another and so on.

The property of a node is that the potential on it is the same. The convention is to mark a node with an alphabet, and then call the potential at that point V_a for instance.

Solution:

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(d) Mark which nodes are 2-nodes and multi-nodes. 2 nodes are connected to only 2 components, and multi-nodes are connected to 3 or more components.
Note: *Whenever nodes are marked across which equivalent resistance must be found, those are considered 'components' because something could be connected there.*

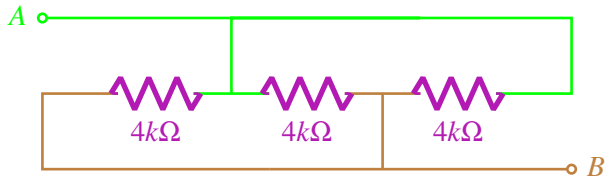
Solution: Multinode and 2-node are not official vocabulary, just aids

Solution: Nodes A and B are multi-nodes. Nodes C and D are 2-nodes.

(e) Resistors that are connected to 2-nodes are considered to be in series. Redraw the circuit, and find the 2 and multi-nodes again after combining the resistors connected to 2 nodes.

Solution:

Nodes C and D disappear and become nodes A and B.



Solution: Mentors please explain how to combine resistors in series. The simple idea is that since they are on a 2-node, you can drag one till you reach the other, and just write $R_1 + R_2$ on that.

(f) Now we should be left with only multi-nodes. So far, we have seen that any resistors connected to 2-nodes are in series. We will see what happens to resistors connected to multi-nodes. Begin by writing out the 2 nodes that each resistor is connected to.

Solution: The resistor on the left is connected to nodes A and B. The resistor on the middle is connected to nodes A and B. The resistor on the right is connected to nodes A and B.

(g) If you have 2 or more resistors that are connected to the same 2 nodes, then they are in parallel. What does this mean for the 3 remaining resistors?

Solution: Since all 3 resistors are connected to nodes A and B, they are all in parallel. The equivalent resistance is $\frac{4}{3}k\Omega$. Quick tip: if you have 2 resistors in parallel, both of whose resistance is R, then the equivalent resistance is $\frac{R}{2}$. Similarly, if you have 3 resistors in parallel, all three of whose resistance is R, then the equivalent resistance is $\frac{R}{3}$, and so on with 4 or more resistors of the same value.

Solution: If students don't realize the tip, show them by deriving it with the formula.

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3. Resistance is not Futile!

Learning Goal: Introduce the concept of resistivity and provide physical intuition for how changes in physical properties/dimensions of a resistor changes its resistivity. Read *Note 12 Section 12.3-12.4* to learn more.

Meta:

- Having one consistent analogy throughout the question, such as a water pipe, can help give students a better mental image of what changes between parts.
- Relate putting wires lengthwise to resistors in series and putting wires side to side to resistors in parallel.

Resistivity is a **physical property** of the material that quantifies how much it opposes the flow of electric current.

Assume that in an ideal case, the cross-section and physical composition of the wire are uniform, We can find its resistance with the equation below:

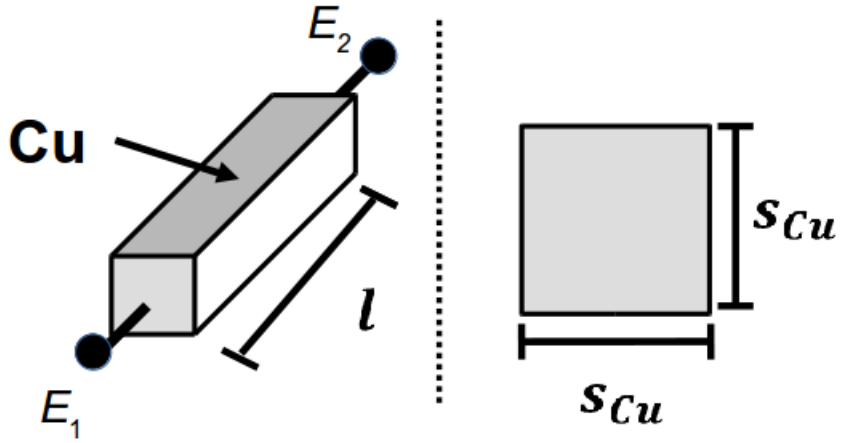
$$R = \rho \frac{L}{A}$$

Here, ρ stands for the resistivity of the wire, R stands for its resistance, A stands for the area of the cross section of the wire, and L stands for the length of the wire.

We will be frequently referencing some of the following variables:

- A: the cross section area of a single wire.
- L: the length of a single wire.
- ρ_{cu}: resistivity for the material copper.
- ρ_{Al}: resistivity for the material aluminum.

(a) A copper (Cu) structure with a square cross-section is shown below. Given the material parameters, calculate the resistance R_{Cu} of the structure between E_1 and E_2.



ρ_{Cu}	$1 \times 10^{-8}\,\Omega\,\text{m}$
s_{Cu}	5 nm
l	75 nm

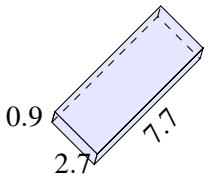
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Solution:

Using the formula for resistance R, we can find the resistance of the Cu structure given the resistivity and the dimensions:

$$R = \rho \frac{l}{A} \implies R_{\text{Cu}} = \rho_{\text{Cu}} \frac{l}{s_{\text{Cu}}^2} = 1 \times 10^{-8}\,\Omega\,\text{m} \cdot \frac{75 \times 10^{-9}\,\text{m}}{\left(5 \times 10^{-9}\,\text{m}\right)^2} = 30\,\Omega$$

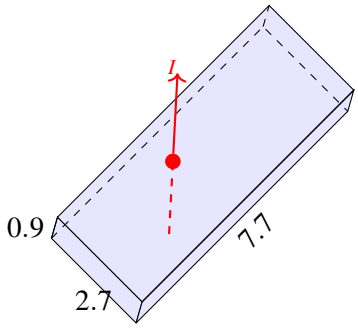
(b) Now you are given a copper slab with dimensions 0.9cm, 2.7cm, and 7.7cm as denoted on the figure below. The dimensions and the resistivity of the slab remain the same throughout the rest of the problem.



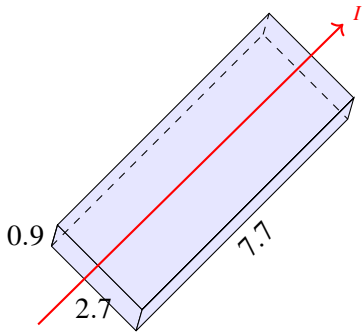
Suppose we connect opposite faces of the slab to a voltage source so that current can flow through it. Notice that there are **3 possible configurations** we can have for this slab, leading to 3 possible directions in which the current can flow (drawn in the answer choices below). Which direction will lead to the **highest** current flow I assuming we use the same voltage source all three times?
Assume the resistivity is the same throughout the slab and does not vary with respect to

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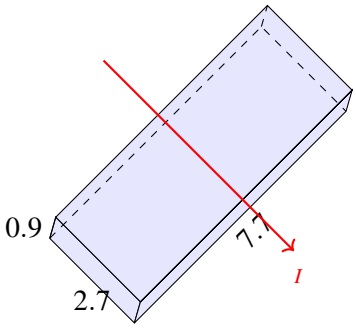
i. Parallel to 0.9 (current goes through 2.7, 7.7 face):



ii. Parallel to 7.7 (current goes through 0.9, 2.7 face):



iii. Parallel to 2.7 (current goes through 0.9, 7.7 face):



iv. Choices (i) and (iii) are tied for the maximum current flow.

v. The direction doesn't matter; the resistor is physically the same, so the current will be the same.

Meta: Emphasize/mention the different cross sectional area and length for each corresponding figure, and try to tie things back to Ohm's law.

Solution: For a given supply voltage from a voltage source, the most current will flow when the resistance of the slab is smallest. We apply the geometrical resistance formula, $R = \frac{\rho_{Cu} L}{A}$. To maximize current, we need to minimize resistance, which requires minimizing L and maximizing A. Two of the dimensions will comprise the area, and the third will form the length. Noticing that $0.9\text{cm} < 2.7\text{cm} < 7.7\text{cm}$, it follows that we want to feed current into the $2.7\text{cm} \times 7.7\text{cm}$ face and use 0.9cm as the length. Therefore, we select choice (i).

(c) Suppose we have N wires similar to the one described in part (a). We align them side by side to form a bundle of wires. Find the overall resistance of this bundle. Are the wires connected in series or parallel?

Solution: Since we have all N wires aligned side by side, we are essentially expanding the cross-section area. This means that the new bundle will have a new cross-section area of NA, while its length remains the same (L). Hence, the overall resistance of the mega-wire will be:

$$R_{mega} = \rho_{cu} \frac{L}{NA}$$

This is similar to a parallel combination of N wires. Note that the equivalent resistance is smaller for a parallel combination.

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(d) **(PRACTICE)** Again how can we connect these N identical wires so that the equivalent resistance is the highest?

Meta: In the next two questions, have students derive the equations for series and parallel resistors.

Solution: The key of this question is to start from the resistance equation:

$$R = \rho \frac{l}{A}$$

Algebraically, we want to maximize the value of R for this question. Since ρ is just a physical constant, we can't changes its value. Observing the fraction $\frac{l}{A}$, we can see that the overall length of the mega-wire should be as great as possible, while its cross section area should be kept as small as possible. How can we arrange N wires in a way so that the overall new wire is as long as possible?
We can arrange the wires in a single long line! This keeps the cross section area A unchanged, but it has a length of NL. Applying the resistance equation, we have:

$$R_{mega} = \rho_{cu} \frac{NL}{A}$$

This configuration is exactly the same as a series connection of resistors. If you think in terms of the equivalent resistance for a series circuit, it also makes sense since we are summing up all the resistances.

(e) **(PRACTICE)** Consider part (c) again, but this time, instead of N copper wires, we split the number evenly between aluminum wires and copper wires. We arrange $N/2$ copper and $N/2$ aluminum wires side by side, and push them to form a new bundle of wires. What is the overall resistance of this wire? (In terms of ρ_{cu}, ρ_{Al}, L, and A)

Solution: As we can see from part (c), when we are aligning the wires side by side, we are essentially arranging the wires to be parallel to each other. We can consider the bundle of wires to be a parallel combination of a bundle of copper wires and a bundle of aluminum wires! For both wires, they will have a length of L and an overall cross section area of $(N/2)A$ (since we have $N/2$ wires for each category). Hence, applying the resistance equation again, we can find that:

$$R_{Cu-bundle} = \rho_{Cu} \frac{L}{\frac{N}{2}A} = \rho_{cu} \frac{2L}{NA}$$

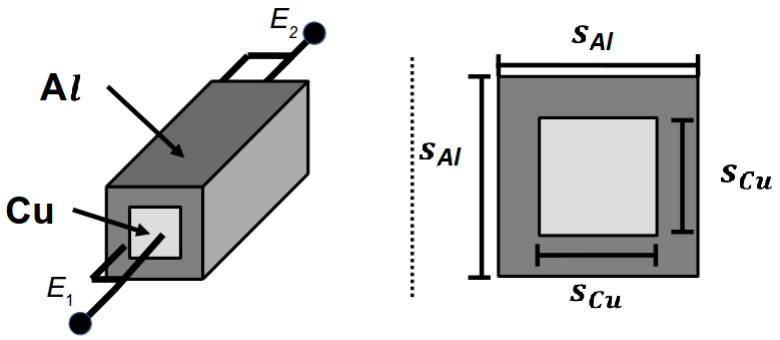
$$R_{Al-bundle} = \rho_{Al} \frac{L}{\frac{N}{2}A} = \rho_{Al} \frac{2L}{NA}$$

Now, since these 2 mega wires are parallel to each other, by equivalent resistance, we can find the overall resistance of the wire to be:

$$R_{overall} = \frac{1}{\frac{1}{R_{Cu-bundle}} + \frac{1}{R_{Al-bundle}}} = \left(\frac{\rho_{Cu}\rho_{Al}}{\rho_{Cu} + \rho_{Al}} \right) \frac{2L}{NA}$$

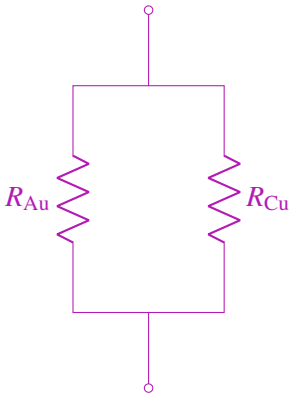
(f) Now consider this *core-shell nanowire* structure, where the outside shell made of Al and the inside made of Cu. E_1 and E_2 terminals are both connected to the faces of the Cu and Al structure, as shown in the picture. Draw a circuit diagram of the nanowire as a set of resistors, using R_{Al} for the resistance of the Al layer and R_{Cu} for the resistance of the Cu layer.

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Solution:

Given that we are contacting the full area and that current is flowing from end to end, each end can be treated as a node since each end will have the Au and Cu at the same potential. This means that we can model the core-shell nanowire as a set of parallel resistors.



(g) Based on your model from part (f) and the parameters given below, find the equivalent resistance R_{wire} between E_1 and E_2.

ρ_{Al}	$2\times10^{-8}\,\Omega\,\text{m}$
s_{Al}	10 nm
ρ_{Cu}	$1\times10^{-8}\,\Omega\,\text{m}$
s_{Cu}	5 nm
l	75 nm

Solution: Using the formula for resistance R, we can find the resistance of the Cu structure given the resistivity and the dimensions:

$$R=\rho\frac{l}{A}\implies R_{\text{Cu}}=\rho_{\text{Cu}}\frac{l}{s_{\text{Cu}}^2}=1\times10^{-8}\,\Omega\,\text{m}\cdot\frac{75\times10^{-9}\,\text{m}}{\left(5\times10^{-9}\,\text{m}\right)^2}=30\,\Omega$$

Using the formula for resistance R, we can find the resistance of the Al structure given the resistivity and the dimensions. In this case, the area of the Al structure is not just a square but rather the area of the Cu square subtracted from the area of the Al square (the area of a shell).

$$R_{\text{Al}}=\rho_{\text{Al}}\frac{l}{s_{\text{Al}}^2-s_{\text{Cu}}^2}=2\times10^{-8}\,\Omega\,\text{m}\cdot\frac{75\times10^{-9}\,\text{m}}{\left(10\times10^{-9}\,\text{m}\right)^2-\left(5\times10^{-9}\,\text{m}\right)^2}=20\,\Omega$$

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Because the two structures are in parallel, the total resistance R_{wire} is:

$$R_{\text{wire}} = R_{\text{Al}} \parallel R_{\text{Cu}} = \frac{20\,\Omega \cdot 30\,\Omega}{20\,\Omega + 30\,\Omega} = \frac{600\,\Omega^2}{50\,\Omega} = 12\,\Omega$$

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